MANAS GIRDHAR

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Education

Birla Institute of Technology, Mesra **Aug 2024 – June 2028**
Bachelor of Technology in Computer Science and Engineering **CGPA: 8.92 / 10.0**

Technical Skills

Languages: C++, JavaScript (ES6+), Python, SQL
Backend & APIs: Node.js, Express.js, FastAPI, Flask, REST APIs, Passport.js (authentication)
Backend Security: Input validation, authorization, password hashing, rate limiting
Frontend: React.js, EJS, HTML, CSS
Databases & SQL: MongoDB, MySQL; joins, indexes, transactions, query optimization
CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Development & Testing: Git, GitHub, GitHub Actions (CI), pytest, Postman, Linux (Bash), Vercel, Render
Applied AI/ML: Scikit-learn, XGBoost, CatBoost, Pandas, NumPy, Sentence Transformers, Gemini API

Technical Projects

Smart Cart Optimization Engine GitHub | Live
React.js, Node.js, Express.js, MongoDB, Sentence Transformers, Gemini API

- Owned the **recommendation component end-to-end** in a team-built MERN application, implementing product retrieval, ranking and recommendation explanations within its 3-service backend.
- Implemented **semantic retrieval** using Sentence Transformer embeddings and cosine similarity to identify relevant products from the available catalog.
- Built a **hybrid scoring pipeline** combining semantic similarity, ratings, price compatibility and popularity through weighted scores to rank recommendations.
- Integrated **Gemini-generated explanations** with a **4-level fallback mechanism**, response validation and error handling to handle failed API calls and malformed model outputs.

GridLock AI – Traffic Intelligence Platform GitHub | Live
React.js, Python, FastAPI, Pandas, NumPy, XGBoost, CatBoost

- Built a React dashboard with **4 operational modules** for routing, incident management, officer deployment and traffic analytics, backed by **6+ FastAPI endpoints** connecting frontend and prediction services.
- Developed a Pandas/NumPy preprocessing pipeline using **K-Means clustering and target encoding** to transform raw city-event records into model-ready features.
- Combined XGBoost, CatBoost and Logistic Regression in a **stacking ensemble** to predict road-closure probability and congestion severity for planned and unplanned city events.
- Implemented an **A*-based route planner** that recommends congestion-aware diversions and exposes travel-time and delay estimates through the application.

HeartRisk 3D – Coronary Risk Visualization Live | Video
Python, Flask, SHAP, pytest, GitHub Actions, GitHub Pages

- Built a **Flask prediction API** and interactive 3D heart for the IIT Mandi Multimodal AI Hackathon 2026, visualizing coronary artery disease risk and predicted narrowing of **3 coronary arteries**.
- Implemented **browser-side inference** using the same model as the backend and parity-tested browser/API outputs to check consistency across both execution environments.
- Added **SHAP explanations and calibrated uncertainty** alongside predictions to expose model behavior and uncertainty alongside the risk visualization.
- Added **9 pytest tests** and GitHub Actions for automated checks; published the browser application on GitHub Pages.

YelpCamp – Full-Stack Marketplace Live
Node.js, Express.js, MongoDB, EJS, Passport.js, Mapbox, Cloudinary

- Built a campground marketplace using Node.js, Express.js and MongoDB with **Passport.js authentication**, server-rendered EJS pages, campground listings and image uploads.
- Integrated **Mapbox geocoding** to map campground locations and **Cloudinary** to store uploaded images and display multi-image galleries; deployed on Render.

Achievements

- **Flipkart GRiDLock Hackathon 2.0 Finalist** (Top 4.7% of 34,600+ registrations); **Flipkart GRiD 8.0 National Semi-Finalist**, advancing to Round 3.
- Solved **700+ algorithmic problems** across LeetCode, Codeforces and GeeksforGeeks, including **400+ Medium/Hard** problems; peak LeetCode contest rating: **1788**.
- Achieved **9.45 SGPA** (Top 5% of the BIT Mesra B.Tech CSE cohort); received the **G. P. Birla Scholarship** for academic merit.